



Merge curve

Absorb one edge into its neighbour as a single fitted spline -- Edit curve / Composite + Approximate in one step -- and keep the right body.



What it does

A short edge next to a long one (a tip section, a sliver left by a split) wants to be part of it. Onshape has no join; every route is fit-then-replace, and doing it by hand loses the wire's identity. Merge curve fits the two edges as one spline and puts it back where it belongs.

Use it when

- Two edges of a wire should be one smooth curve
- The wire's name and identity must survive
- The ends must still meet their neighbours exactly

Merge curve: the RD baseline after merging -- 2 edges left, the merged one in blue (height x20)

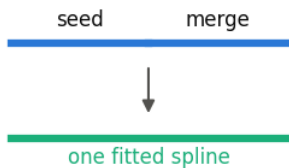


The RD baseline after merging its tip section: 2 edges left.

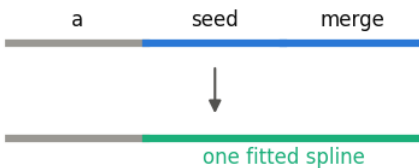


Which body survives

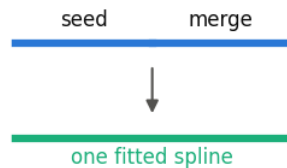
In place
the seed is the only edge of its wire:
the wire is edited, keeps its id and name



Rebuild
the seed wire has other edges:
they + the merged curve become a new wire



Extract
the seed is not on a wire:
the fitted spline is a new wire



In place

The seed is its wire's only edge: the wire is edited and keeps its id and name.

Rebuild

The wire has other edges: they and the merged curve become a new wire (info notice).

Extract

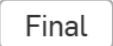
The seed is not on a wire: the fitted spline is the output.



The dialog

Merge curve: RD Baselin...  

- Seed edge
Edge of Derived 1 
- Merge edge
Edge of Derived 1 
- ☐ Keep start derivative
- ☐ Keep end derivative
- > Approximation
- Output name _____
- ☐ Note arc loss
- > Debug

#	Parameter	What it means
1	Seed edge	The edge that is edited.
2	Merge edge	The edge absorbed; it must meet the seed end to end.
3	Keep start derivative	Keep the seed's own start tangent (on).
4	Keep end derivative	Keep the merge edge's own end tangent (on).
5	Approximation	Degree, tolerance, maximum 100 control points.
6	Output name	Pre-filled with the seed wire's name.
7	Note arc loss	Say so when a line or arc becomes a spline.
8	Debug	Print diagnostics.



Example: the RD baseline tip

Merge curve: the RD baseline after merging -- 2 edges left, the merged one in blue (height x20)



The baseline's tip section merged into its main edge (the merged edge in blue)

The seed wire had other edges, so it was rebuilt: the untouched edge and the merged curve form the new wire, the ends snapped to their neighbours so it still chains.



Outputs and messages

Published for Extract variables

Key	What it holds
startVertex / endVertex	The ends of the result (start = the tail end: arc length grows with world X).
startEdge / endEdge	The edge at each end.
mergedEdge	The merged curve's edge.
degree / controlPointCount	Of the fitted spline.

Messages

	Message / what to do
Info	Rebuilt the seed wire ... original body was replaced The Rebuild case.
Info	arc radius not preserved A line or arc became a spline.
Error	not G0 connected The two edges do not meet end to end.



Tips and limits



Always a spline

A merged line or arc loses its type.



Many edges?

To clean a whole wire use Clean wire instead.



Where it lives

Document	Curve_tools (2143812a99089658c704f0bc)
Tab	merge_curve (uses curve_core)
Tests	Tests studio (curve_tools_tests: merge, ends)
Examples	Test_1 studio
Explainer	docs/explainers/curve_tools/curve_tools_explained.md, section 2.3
Related	Clean wire, Map curve, Merge curve, Evaluate profiles, Fillet wire; Extract variables